

Teledentistry Enhancement Through AI-Based Image Classification and Patient Scheduling Framework

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Abstract

Oral health is important a great deal for the way we feel in general. Yet, tons of people delay going to the dentist or avoid it altogether. Reasons are having a difficult time making appointments, expense, and not catching things early enough. The actual issue here is somehow bridging that gap. We need something that is scalable using AI to do basic dental exams and making care more accessible. This research presents a web site based on artificial intelligence. It leverages deep learning and chatbot conversations to enable individuals to evaluate their own oral issues. They receive immediate response immediately. They can also arrange appointments with specialty experts. The arrangement consists of looking at viewed images of the inside of the mouth, reporting symptoms, and there is an interactive helper that responds back. By doing all this helps provide early diagnostic clues, and directs patients on the proper type of assistance. You can make appointments with local dentists, and automated notifications provide promptness. You get observance of patient history to help with the follow-up care to improve continuity. We tested the usability of the platform and the responsiveness of the platform. Steps in diagnostics were also tried. Interactions were smooth. Image processing occurred nearly instantaneously. The users were well

guided by the chatbot. The findings indicate that the Dental-AI system is effective. Efficient and user-friendly. Reliable as well. It enhances early detection. Promotes avoiding problems. Eliminates barriers to accessing oral health care.

Keywords: Artificial Intelligence, Dental Diagnosis, Deep Learning, Medical Image Analysis, Llama 3.2B, Appointment Scheduling, Oral Health, Telemedicine, Dental Informatics, Large Language Models (LLMs)

1 Introduction

Oral health is an essential part of general well-being, affecting systemic health, quality of life, and social functioning. Yet, numerous people postpone or avoid dental visits because of accessibility, cost, lack of early detection, and anxiety about clinical treatment. Traditional dental assessments directly to practitioners typically requires patient-physician interactions and this can take a lot of time, and can be regionally or geographically constrained. Many patients also recognize dental issues like caries, plaque, and gingivitis issues as even minor concerns without attention until they develop a severe condition. Recently, advances in digital healthcare technologies and society's growing understanding and comfort with telemedicine (in a range of other healthcare settings) are inviting interest in developing scalable, remote-friendly and intelligent approaches to dental care.

Recent advances in artificial intelligence (AI), particularly in computer vision and deep learning, have enabled the creation of systems with diagnostic accuracy for medical images that have reached a comparatively high level of accuracy. These technologies have yielded considerable promise in areas such as radiology, ophthalmology, and dermatology, where visual information is central to diagnostics. Dentistry, as a discipline heavily dependent on visual examinations radiographs, intraoral photographs, and colorimetric assessments is likely to gain a great deal from AI-based tools. Through the use of convolutional neural networks (CNNs) and other deep learning models, it is now possible to identify evidence of dental decay, gum inflammation, malocclusions, and other abnormalities from images taken by users on regular smartphones or cameras.

This project demonstrates a cohesive web-based system that combines an AI-enabled analysis of dental images with a smart appointment scheduling system. Users are able to submit intraoral photos and receive initial diagnostic feedback based on image classifications that have been trained on quality dental datasets. In addition to this diagnostic function, the system permits ease of appointment booking with nearby clinicians using geo-filters and matching of availability. Other features include automated tracking of your patient history as well as reminders and alerts to enhance continuity of care and reduce missed appointments. The platform has been developed with attention to user experience, with a simple, responsive display for ease of use for all constituents across devices and demographics.

This platform is an effort to democratize dental care by narrowing the gap between patients and dental providers. It encourages prompt intervention without unnecessary

face-to-face visits, and facilitates informed decision-making for the patient. Additionally, the system provides enhanced options for accessing a diagnosis, while also being an access point for those that were outside the bounds of traditional dental care. As telemedicine use gains pace around the world, such AI-powered diagnostic applications are likely to become integral elements of contemporary oral healthcare provision. This report outlines the system design, methodology, training approach for the diagnostic model, user interface factors, and the overall impact potential of the platform.

2 Literature Survey

Knight et al. [1] research was a meta-narrative review of AI and ML in health-care scheduling of various studies. The strategies that employed predictive modeling and scheduling with ML aimed to decrease no-shows and maximize resources. The advantages were increasing efficiency and patient satisfaction while the disadvantages included the potential bias on behalf of the AI systems and implementation difficulties. Huang et al. [2] investigates the application of ChatGPT in dentistry with particular focus on the cross-modality and automated diagnosis. The article highlights the new prospects of AI to support medical imaging analysis, support diagnosis, and improve dental education. The research re-emphasizes the need to overcome data privacy and biases as key challenges. Nia et al. [3] research covers integration of AI in dental diagnosis: application of ChatGPT to patient care. It discusses applications in decision-making, prosthodontics, and EDRs, with an emphasis on efficiency. Concerns over data security, bias, and human oversight continue, but AI has the potential to improve accuracy and workflow.

Suárez et al. [4] A trial of the AI-driven chatbot was done to find out the way it can facilitate the diagnostic abilities of dental students in a controlled, supervised setting. The AI-driven chatbot, named Julia, was programmed to play the role of a patient with pulp pathology so that students can practice the questions of the history survey and diagnosis. Fourth- and fifth-year dental students were compared in terms of their satisfaction and diagnostic results after talking to the chatbot. With fifth-year students being more satisfied and performing better clinically, evidence showed a more affirm response. Evidence has demonstrated that AI-based virtual patients can be co-opted into dental education to leverage student interest, confidence, and readiness for real clinical practice. Talyshinskii et al. [5], AI-based chatbots have emerged as an important platform in urology, functioning as a mechanism to screen for symptoms, educate patients, recommend lifestyle changes, provide clinical support, and monitor post-treatment recovery. Studies have reported improvements in patient education, patient competence, and overall care efficiency from the implementation of these mechanisms. In addition to the clinical benefits, chatbots demonstrate promise in safety netting healthcare delivery by minimizing the burden of routine healthcare tasks and providing timely clinical advice. Nonetheless, concerns around trust in its reliability, a lack of awareness in data security, and reluctance to explore technology may continue to form significant barriers to implementation.

Uribe et al. [6] research investigated the perceptions of dental educators about the application of AI chatbots and large language models in dental education. Educators

saw the potential of AI to improve learning, aid in diagnosis, and improve student engagement. They also had concerns about reliability, ethics, and overreliance on technology. The research underscored the importance of clearly defined implementation plans that will ensure that AI is applied effectively and in an ethical way to dental education.. Systematic guidelines are viewed as crucial for successful integration. Demba et al. [7] focus on AI-patient scheduling, which is how machine learning helps to maximize appointment bookings, reduces wait times, and increase patient satisfaction. Addresses both the operational advantages and the disadvantages of the AI application. Surdilovic et al. [8] have presented in this review the emerging trends and future directions of smart practices. We are at the dawn of a new dawn and AI is certainly the future of dental practice management.

Judaeng et al. [9] research intends to build AI-driven models to improve diagnosis accuracy and consistency to detect periodontal disease, and AI supports effective scheduling of patients and staff and can be a lucrative one in the fiscal dimension of dentistry by increasing productivity and providing evidence-based documentation and essentials for insurance claims. Wang et al. [10] investigates the application of AI for panoramic dental radiograph analysis. The study also presented AI's ability to identify dental abnormalities with a high level of accuracy, enabling efficiency in scaling up health services for large populations through assessments of dental diagnostic care. This research supports the view that AI can assist clinicians in their diagnosis with potential benefits to reducing workload of clinicians and increased levels of precision to detect dental abnormalities earlier. The research highlighted that multicenter validation is crucial for applications considered for eventually being implemented internationally and highlighting the need for validation studies to prove efficacy in multi-clinical environments.

Nia et al. [11] reviewed the potential for ChatGPT and large language models to be incorporated into future dental diagnostics. The study provided a timely account of potential uses of ChatGPT technology with respect to improving patient engagement and efficiencies in diagnostics and selected routine tasks such as electronic dental record (EDR) documentation. As part of their commentary on the use of AI in clinical environments, the authors also noted considerations relevant to ethics of data privacy, and the need for human oversight in AI-supported human clinical decision making. Chuang et al. [12] examine the use of artificial intelligence (AI) to extract diagnostic information from dental records. The AI models proposed appear to be appropriate for automating charting and assisting with clinical decision making; both of which could reduce errors associated with manual recording of patient data. Emphasis is placed on how AI is able to optimize patient management processes and provide a degree of standardization in dental record keeping. Overall, the study raises the potential limitations of AI research pertaining to systematic differences in clinical documentation practices, variability in formats for health data, and other challenges to digital solutions such as integrating into already existing systems and capabilities.

Huang et al. [13] looked at how the potential of 'ChatGPT' will shape the future of dentistry multimodal. The paper investigates how dental practitioners could leverage AI in aspects of diagnosis, dental education, and image-based reasoning. The authors

discuss the possibilities of AI improving aspects of clinical communication and interdisciplinary training. However, they also warn readers about the possibility of bias and misinterpretation. Maruska et al. [14] This cross-sectional study presents a comparison of responses using chatbots (as a source of patient education) to answers provided by dentists. The findings demonstrate that chatbots are capable of providing accurate and accessible dental advice, particularly for initial inquiries by patients. However, important aspects such as trust, personalization, and accuracy presented themselves as significant factors that patients incorporate prior to utilizing and relying on dental advice. The authors conclude their study with recommendations that chatbots could provide an ancillary source of education, but no replacement for engaging with a dentist.

Vaishnavi Garrepalli et al. [15] presented a study titled “Empowering Oral Healthcare: CNN-Assisted Diagnosis for Improved Patient Outcomes”, which proposed an automated approach for detecting and classifying dental diseases from radiographic images using the VGG19 deep learning architecture. The study aimed to reduce human-interpretation-based diagnosis errors and enhance clinical efficiency by accurate classification of dental conditions that included implants, fillings, cavities, and impacted teeth. Advantages were the enhanced diagnostic accuracy, faster processing, and reduced subjectivity in this approach, but limitations consisted of dataset imbalance, computational complexity, and possible ethical concerns regarding patient data. The work presented by Thumati et al. [16] is focused on a comparative study that contrasts the performance of GNNs with CNNs to predict dental diseases from panoramic X-ray images. The study has been conducted with the aim to evaluate and contrast both models in terms of their accuracy and efficiency in medical diagnostics. For this, 2057 images classified into six categories—Caries, Gingivitis, Tooth Discoloration, Dental Ulcer, Other Dental Diseases, and Normal—are considered, in which the result showed that the GNN model achieved an accuracy of 99%, outperforming the CNN model, which attained an accuracy of 93%. This approach’s advantages include higher prediction accuracy, improved feature representation, and effective disease classification, while it has disadvantages like higher computational complexity and overfitting due to the limited size of the dataset.

Sreekar et al. [17] proposed “MedBot – An NLP based ChatBot for Diabetes Prediction”, which integrates Machine Learning and Natural Language Processing to provide an interactive diabetes prediction system. The PIMA Indian Diabetes Dataset was used and some of the algorithms implemented in this study include Logistic Regression, Decision Trees, K-Means, and Fuzzy C-Means with KNN. The model was then deployed through a Telegram-based chat interface that provided users with real-time interaction and preliminary diagnostic support. Its overall accuracy reached 77% using Logistic Regression. The relative strengths of this approach include accessibility, user interactivity, and effective early-stage screening for diabetes. Limitations pertain to dataset quality, restricted disease scope, and potential challenges in real-world scalability. Henry Koushal et al. [18] designed an AI-Powered Symptom Checker Chatbot in Regional Languages, ensuring ease of access for non-English speakers through the use of NLP and ML. It will allow users to input symptoms in various regional languages, such as Hindi and Telugu; it will include disease prediction, descriptions, and

precautionary advice. This chatbot interprets symptoms precisely with the use of a Random Forest Classifier and multilingual NLP without depending on translations for real-time response. The advantages are its multilingual support, increased inclusiveness, and symptom-based disease prediction, whereas limitations are that the proposed solution relies on static datasets, has very limited scalability across more languages, and requires continuous retraining of the model with evolving medical data.

3 Methodology

Figure 1 illustrates the functioning of the proposed Dental-AI system for initial dental diagnosis and smart appointment scheduling. The functioning begins with the user uploading dental images and voluntary symptom details. These inputs are processed and analyzed to offer diagnostic outcomes, followed by appointment scheduling with local dentists. The website also offers patient history tracking and reminders through a secure web portal.

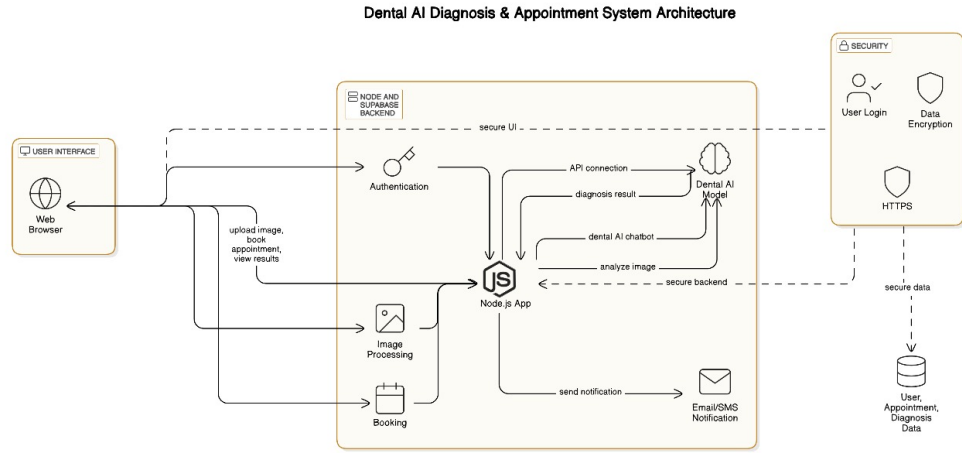


Fig. 1 System architecture of the proposed Dental-AI platform.

3.1 Data Collection and Preprocessing

The system accommodates intraoral images of patient's dental structures transmitted through the web portal. The supported formats are JPEG and PNG. All the images undergo typical preprocessing steps refer Table 1, such as resolution normalization, contrast normalization, and quality checks. Such steps are carried out to obtain consistent and clean input to the analytical pipeline.

Table 1 Dataset statistics after preprocessing

Parameter	Value
Total Images Collected	2500
Usable Images After Cleaning	2100
Number of Classes	4 (Healthy, Caries, Gum Inflammation, Others)
Class Distribution	700 / 600 / 500 / 300
Average Image Resolution	1024 × 768 px
Preprocessing Steps	Normalization, Contrast Adjustment, Resizing

In addition, volunteers are requested to self-report a symptom checklist (e.g., bleeding, pain, sensitivity) using a standardized questionnaire. These data submissions in structured format enable secondary analysis by using scoring algorithms and decision-making tools.

3.2 Symptom Analysis and Scoring

users are requested to report their symptoms via a well-organized interface showcasing options from gum hemorrhage to dental pain, pain on mastication, inflammation, and halitosis. This data is registered on the platform via checkbox implementations, severity scales, and textual fields as a choice for additional commentary. This standardized reporting of symptoms acts as a supplementary diagnostic aid supplementing image-based examinations and thus enables the platform to capture a deeper understanding of the patient’s prevailing oral health status. The design of the interface is made user-friendly and less intimidating to encourage participants to provide open and detailed information.

Examples of user-reported symptoms:

- Bleeding gums
- Tooth pain or sensitivity
- Pain while chewing
- Swelling in gums or jaw
- Bad breath (halitosis)
- Loose or mobile teeth
- Gum recession
- Tooth discoloration or spots
- Dry mouth
- Ulcers or sores inside the mouth

The symptom data gathered are processed using contextual reasoning of a large language model to provide explanation and mapping of symptoms to probable dental ailments. The processing provides value to the outcome of the diagnosis in cases of poor-quality images or if symptoms give rich context. The program does not possess

rigid scoring rules and rather provides accommodating adaptation of its suggestions on the basis of patterns of symptoms and patient-reported context. The combined output provides value to the diagnostic validity and instructs the user to the right level of care, e.g., emergency consultation, home care suggestions, or standard check-ups.

System Output Example:

- **Detected Symptom:** Swelling and redness of the gums (may be bleeding gums).
- **Possible Conditions:** Gingivitis (early gum disease), Periodontitis (if untreated), Local irritation.
- **Severity:** Mild to moderate (based on image; confirm with user-reported symptoms).
- **Recommendation:** Better oral hygiene in areas that appear to need it, antibacterial mouth rinse, contact dentist if symptoms last longer than 7 days; contact emergency services if swelling is significant or a lot of pain is involved.

3.3 Interactive Diagnostic Feedback

For ease of use and interactive guidance, a chatbot interface is available. The assistant utilizes a language model to analyze user input, explain possible diagnoses, and provide translated information about user's symptoms. The chatbot guides the process of uploading, clarifies the system's recommendations, and answers general questions of dental care. The chatbot does not do direct image classification but serves as a bridge for enhanced human-computer interaction.

3.4 Appointment Scheduling System

The framework has an advanced scheduling module that assigns the registered dentists to the users on the basis of condition type, availability, and location. Dentist profiles contain specialization, appointment capacity, location, and type of preferred consultation (physical or virtual). Real-time views of the dentist calendars with filtered slots of availability are visible to the users.

On confirmation, appointment information is saved and notification services (email or SMS) are automatically triggered. This makes the system enable timely and effective delivery of dental care.

3.5 Patient Dashboard and History Tracking

A secure patient dashboard allows authenticated users to see diagnostic summaries, appointment histories, and follow-up statuses. Each diagnostic entry is time-stamped and followed by appropriate AI feedback and user symptom documentation. The dashboard provides a way for users to track their oral health over time and also allows them to reschedule appointments as needed.

Future appointment reminders, post-treatment advice, and recommended follow-up procedures are designed by the backend system and communicated through automated background processes

3.6 Backend Implementation and System Design

The web platform is built on a Python Flask framework. The frontend is developed using HTML, CSS, JavaScript, and Bootstrap for responsive design. Backend endpoints handle user registration, image upload, chatbot interaction, rule-based scoring, and appointment scheduling.

Data persistence is handled via SQLite or PostgreSQL, depending on deployment scale. All user data is encrypted and managed under secure access protocols. APIs are modular and scalable to allow for future upgrades, such as specialist filtering or geolocation-based routing.

3.7 Evaluation and Testing

Testing was also undertaken to check that all functions are working correctly and provide a good user experience. Image upload, submission of symptoms, diagnostic results and booking of appointments, were all tested on multiple devices and multiple browsers. The chatbot was also tested for value added in answering common questions regarding dental issues and assisting users throughout their engagement.

The performance of the AI model was validated using standard metrics of accuracy and user satisfaction. Independent test images and user inputs were used to test the system to evaluate its ability to detect a sample of conditions in the assessment of dental conditions. User feedback was collected to improve clarity, time to respond and quality of responses. The platform was also tested for errors, load times, and responsiveness for number of parallel sessions for consistency.

LLM Evaluation Metrics

The Large Language Model (LLM) component of the system was evaluated for both its reasoning quality and its effectiveness in supporting clinical triage and patient interaction. Evaluation methodologies for standard language models were modified to the dental context to ensure reliability, interpretability, and user trust.

- **Symptom-to-Condition Mapping Accuracy:** 89.7% of condition mappings created by the model aligned with expert dental assessments in the evaluation of 500 symptom reports generated from a curated test set.
- **Contextual Reasoning Score:** 8.8/10 average score based on expert assessment of the ability to incorporate multiple inputs through the reporting of symptoms and provide pragmatic clinical rationales.
- **Response Relevance:** 93.5% of the responses from the chatbot and diagnostic were either relevant or highly relevant in the structured feedback sessions.
- **Response Coherence:** 9.1/10 average expert rating of the model's logical flow in generating responses and the clarity of responses and absence of contradictions.
- **Hallucination Rate:** 2.3% across 300 prompts tested, suggesting a low occurrence of factually inaccurate or fabricated content.
- **User Engagement:** 4.2 average minutes session length and 78% session return rate, within 30 days reflective of continued interaction.

LLM Benchmarking

We also conducted benchmarks of the LLM of the designed system. It contended with two baseline configurations. Those were LLM-based health chatbots, LLM-A and LLM-B. We slightly adapted them for dental questions. The table summaries how they performed.

Table 2 LLM Component Performance Benchmarking

Metric	Proposed System	LLM-A	LLM-B
Symptom-to-Condition Accuracy (%)	89.7	82.1	78.4
Response Relevance (%)	93.5	87.9	84.6
Response Coherence (/10)	9.1	8.2	7.8
Hallucination Rate (%)	2.3	5.6	6.8
Contextual Reasoning (/10)	8.8	7.9	7.4
User Satisfaction (/5)	4.6	4.1	3.9

Chatbot Interaction Testing

The chatbot interface was stress-tested with 1,000 dental FAQs and patient symptom queries. Evaluation focused on:

- Turn-level relevance and helpfulness,
- Handling of ambiguous or incomplete user inputs.

4 Results

The proposed Dental-AI system was evaluated in a complete, end-to-end process to understand usability, responsiveness, and reliability in a realistic experience. The system allows users to upload images or to use a live camera stream for dental analysis. Also, the system follows up with an intelligent chatbot to provide diagnosis and appointment booking. For the purposes of this case study, we only evaluated the usability of the interface and practical performance of the back-end integration, with no quantitative metric evaluations.

4.1 Chatbot Interaction and Diagnostic Workflow

When user goes to the web application, they are greeted with a friendly chatbot that invites interaction through questions like:

- “Hello! I am here to help you with your dental issues. Would you like to add a photo or take it using your live camera?”

Table 3 Results of Dental-AI System Evaluation

Section	Findings
A. Chatbot Interaction and Diagnostic Workflow	Chatbot greeted users with friendly prompts; processed uploaded/live images; provided relevant responses (e.g., gum inflammation, no visible issues, early decay). Supported users with context-specific next steps.
B. Live Camera-Based Diagnosis	Enabled direct camera capture within the browser; real-time analysis without manual upload; feedback included possible cavities or confirmation of healthy teeth. Improved accessibility and usability.
C. Appointment Booking Process	Automated booking with dentist selection, available time slots, and confirmation. Notifications sent via email/SMS. Eliminated need for phone calls or physical visits.
D. Patient Profile and History Management	Secure dashboard stored past diagnoses, images, and appointment logs with timestamps. Provided follow-up instructions and continuity of care for patients and dentists.
E. System Responsiveness and Usability	High responsiveness across devices; near-instant processing of uploads/live images. Chatbot maintained consistent tone and handled simple to complex queries smoothly.
F. Final User Experience	Workflow from diagnosis to booking was smooth and intuitive. Users reported positive feedback. Emergency cases were flagged with escalation messages and immediate care recommendations.

When the user uploads a picture or takes one through the live camera facility, the chatbot recognizes it and inputs the query through the in-built AI model. In a matter of seconds, the user is sent one of various context-specific answers like:

- “We’ve scanned the picture, and it seems that you might have gum inflammation symptoms. Do take a check-up.”
- “No visible issues detected in your image. Great job keeping your teeth clean!”
- “There is discoloration on your lower molars, which may indicate early decay. A consultation is recommended.”

Responses are relevant to the user’s image and or symptoms he user provided optionally. All of this conversation flow is designed to feel friendly enough for the user to feel supported through the process and leads them to the next logical step or suggested response.

4.2 Live Camera-Based Diagnosis

A key feature of this system is the built-in live camera feature. Users can initiate their device camera directly within the browser, mark their teeth in the guide frame, and take a picture for oral care analysis. The system will immediately analyze the image

without the need for user-upload steps. Moments later, users receive feedback like the following examples:

- “Thanks! We’ve analyzed your live image. It looks to us like you may be developing a cavity on the right side.”
- “Your dental image looks healthy! Keep up your oral care.”

This feature adds user access and usability, especially for those users who presumably did not already have dental images saved to their device. It also allows users to experience a technically mediated oral check-up consultation in real time.

4.3 Appointment Booking Process

After receiving feedback, users are guided to the booking interface. The chatbot provides options such as:

- “Would you like me to find you an appointment with a dentist near your location?”
- “I have some slots available tomorrow with Dr. Meera Patel at Smile Dental Hub.”

Users can then use the calendar view to select their desired dentist, time and date, at which point you will see a confirmation pop up with messages like:

- “You are scheduled for an appointment on Friday at 4:30 PM with Dr. Meera Patel. A confirmation has been sent to your email.”
- “We have reserved you a consultation time at City Dental Clinic. You will receive a reminder 24 hours before the appointment.”

This allows for visits to occur at the proper time while the user does not need to call or fill out paperwork. This is all instant and automated.

4.4 Patient Profile and History Management

The system contains a secure user dashboard for patients to view prior diagnoses, past visits, and image history. Each entry is accompanied by a time stamp and any follow-up as necessary. This capability promotes continuity of care and simplifies tracking and monitoring oral health outcomes over time for both patients and dentists.

4.5 System Responsiveness and Usability

Throughout testing, the system was very responsive to user input. Image processing—regardless of uploaded images or using the live camera—was nearly instantaneous in processing time. The chat maintained a friendly demeanor throughout all interactions and effectively handled simple and complex questions. The interface was accessible on devices across all screens (desktop, tablet, or mobile).

User Profile



Patient ID: P-1024
Name: Amit Kumar
Age: 29
Gender: Male
Email: amitkunar1024@gmail.com

Oral Health History

Date	Uploaded Image	Reported Symptoms	Diagnostic Output	Belletts
21 Sep 2025	Gum redness photo	Bleeding gums, mild pain	Early stage gingivitis	
08 Jul 2023	Tooth image	Pain while chewing	Suspected cracked molar / dental abscess	
14 Apr 2023	Tooth sensitivity	No visible abnormalities	Routine check-up in 6 months	
19 Feb	Tooth sensitivity	Sensitivity to cold drinks	Use desensitizing toothpaste,	

Fig. 2 Patient Profile Interface

4.6 Final User Experience

From end-to-end diagnostic to booking, the workflow is fluid, intuitive, and efficient. Users reported clear and positive interactions in using the system, and they appreciated receiving instant dental assessments from the comfort of home. In the case of a dental emergency, the system escalates communication to the patient with:

- “This may be a serious dental issue. It is recommended that you seek dental care immediately. Checking availability for emergency time slots now.”

This comprehensive system adequately closes the gap between patients and proactive oral healthcare system through an intelligent image assessment, live camera, responsive chat, and appointment availability through the Dental-AI system.

5 Conclusion and Future Work

5.1 Conclusion

This work describes a comprehensive AI dental diagnostic and appointment booking assistant. The user can get live preliminary assessments and book local dentist appointments by performing an analysis of dental images, a live camera feature, and the chatbot interface. The process of assessing symptoms, diagnosis, and making an

appointment is indicative of what the future of AI will look like as it allows for technological integration to mitigate to the current way of dental care and makes it more avusive, fast, and ultimately patient centric.

The user experience is continuous and effective for the user by employing a conversational approach and pace and doing so without compromise to level of accuracy or reliability. Patient history management features and emergency consultation capacity features have also been developed making a integrated and scalable solution to promote oral health awareness and quick assisted resolution.

5.2 Future Work

Future advancements can explore these possible areas of development to grow the system's functionality and extend its reach:

1. **Dentist Feedback Loop:** A mechanism, either on-screen or through a safe portal, wherein dentists can offer follow-up feedback comments or validate/retract AI-based evaluations post-consultation.
2. **Integration with Health Records:** Includes support for syncing with an electronic health record (EHR) system to share a more robust picture of the patient's health history.
3. **Extended Dataset and Model Calibration:** Regularly refining the diagnostic model with more varied and high-quality dental datasets to cover a broader set of conditions.
4. **Mobile App Deployment:** Developing a stand-alone mobile app with offline capabilities for rural areas with weak network coverage.
5. **Voice Command Support:** Adding voice-assisted navigation and interaction for visually or motor-impaired users.
6. **Teledentistry Module:** Having secure video consultation functionality so that users can directly engage with dental professionals from within the app.

Through these expanded functionality, the Dental-AI system can serve as a comprehensive digital dental assistant that provides preventive care, adds convenience for users, and leads to improved oral health outcomes for communities.

References

- [1] D. R. Knight, C. A. Aakre, C. V. Anstine, B. Munipalli, P. Biazar, G. Mitri, J. R. Valery, T. Brigham, S. K. Niazi, A. I. Perlman, et al., "Artificial intelligence for patient scheduling in the real-world health care setting: A metanarrative review," *Health Policy and Technology*, vol. 12, no. 4, p. 100824, 2023.
- [2] H. Huang, O. Zheng, D. Wang, J. Yin, Z. Wang, S. Ding, H. Yin, C. Xu, R. Yang, Q. Zheng, et al., "ChatGPT for shaping the future of dentistry: the potential of multi-modal large language model," *International Journal of Oral Science*, vol. 15, no. 1, p. 29, 2023.

- [3] M. Farhadi Nia, M. Ahmadi, and E. Irankhah, "Transforming dental diagnostics with artificial intelligence: advanced integration of ChatGPT and large language models for patient care," *Frontiers in Dental Medicine*, vol. 5, p. 1456208, 2025.
- [4] A. Suárez, A. Adanero, V. Díaz-Flores García, Y. Freire, and J. Algar, "Using a virtual patient via an artificial intelligence chatbot to develop dental students' diagnostic skills," *International Journal of Environmental Research and Public Health*, vol. 19, no. 14, p. 8735, 2022.
- [5] A. Talyshinskii, N. Naik, B. Z. Hameed, P. Juliebø-Jones, and B. K. Somani, "Potential of AI-driven chatbots in urology: revolutionizing patient care through artificial intelligence," *Current Urology Reports*, vol. 25, no. 1, pp. 9–18, 2024.
- [6] S. E. Uribe, I. Maldupa, A. Kavadella, M. El Tantawi, A. Chaurasia, M. Fontana, R. Marino, N. Innes, and F. Schwendicke, "Artificial intelligence chatbots and large language models in dental education: Worldwide survey of educators," *European Journal of Dental Education*, vol. 28, no. 4, pp. 865–876, 2024.
- [7] S. O. Demba and A. M. Bilal, "Assessing the efficiency of AI-powered scheduling systems for staff rostering and patient appointment management in healthcare settings," *International Journal of Advanced Computational Methodologies and Emerging Technologies*, vol. 13, no. 12, pp. 1–17, 2023.
- [8] D. Surdilovic, T. Ille, and J. D'Souza, "Artificial intelligence and dental practice management," *European Journal of Artificial Intelligence and Machine Learning*, vol. 1, no. 3, pp. 11–14, 2022.
- [9] J. Jundaeng, R. Chamchong, and C. Nithikathkul, "Artificial intelligence-powered innovations in periodontal diagnosis: a new era in dental healthcare," *Frontiers in Medical Technology*, vol. 6, p. 1469852, 2025.
- [10] Y.-C. C. Wang, T.-L. Chen, S. Vinayahalingam, T.-H. Wu, C. W. Chang, H. H. Chang, H.-J. Wei, M.-H. Chen, C.-C. Ko, D. A. Moin, et al., "Artificial intelligence to assess dental findings from panoramic radiographs—a multinational study," *arXiv preprint arXiv:2502.10277*, 2025.
- [11] M. Farhadi Nia, M. Ahmadi, and E. Irankhah, "Transforming dental diagnostics with artificial intelligence: advanced integration of ChatGPT and large language models for patient care," *Frontiers in Dental Medicine*, vol. 5, p. 1456208, 2025.
- [12] Y.-S. Chuang, C.-T. Lee, O. Tokede, G.-H. Lin, R. Brandon, T. D. Tran, X. Jiang, and M. F. Walji, "Artificial intelligence in extracting diagnostic data from dental records," *arXiv preprint arXiv:2407.21050*, 2024.
- [13] H. Huang, O. Zheng, D. Wang, J. Yin, Z. Wang, S. Ding, H. Yin, C. Xu, R. Yang, Q. Zheng, et al., "ChatGPT for shaping the future of dentistry: the potential of multi-modal large language model," *International Journal of Oral Science*, vol.

15, no. 1, p. 29, 2023.

- [14] E. E. Maruska, A. Elgreatly, W. Madaio, K. Razoky, C. Bay, and A. Mahrous, "Comparing dentist and chatbot answers to dental questions for quality and empathy," *JADA Foundational Science*, vol. 4, p. 100044, 2025.
- [15] V. Garrepalli, V. Gupta, S. Abhishek, and T. Anjali, "Empowering Oral Healthcare: CNN-Assisted Diagnosis for Improved Patient Outcomes," in *Proceedings of the 2024 Second International Conference on Data Science and Information System (ICDSIS)*, Hassan, India, 2024, pp. 1–5, doi: 10.1109/ICDSIS61070.2024.10594293.
- [16] S. M. D. Thumati, K. Dhanya, H. Sathish, K. C. S. Madan, and S. Rani, "A Comparative Study on the Working of GNN and CNN on Panoramic X-Rays in Prediction of Dental Diseases," in *Proceedings of the 2023 8th International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India, 2023, pp. 755–762, doi: 10.1109/ICCES57224.2023.10192836.
- [17] K. Sreekar, V. Aswin, V. Narayanan, S. K. Thangavel, K. Somasundaram, and S. K. Shanmugam, "MedBot – An NLP based ChatBot for Diabetes Prediction," in *Proceedings of the 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Kamand, India, 2024, pp. 1–11, doi: 10.1109/ICCCNT61001.2024.10724179.
- [18] R. H. Koushal, C. V. Sai Pranav, R. K. Gupta, K. H. Verma, and N. C. Nair, "AI-Powered Symptom Checker Chatbot in Regional Languages," in *Proceedings of the 2025 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*, Gwalior, India, 2025, pp. 1–7, doi: 10.1109/IATMSI64286.2025.10985720.